AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION 2010

Biology Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL NINE questions. Questions 7, 8 and 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

Scientific advances in technology have a notable impact on human welfare. Justify the given statement using cloning as an example.

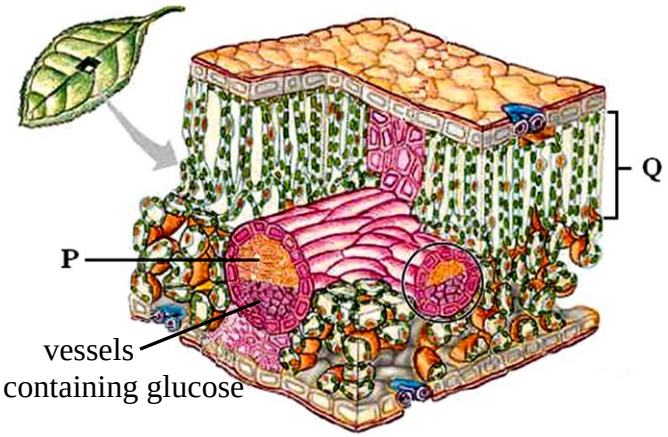
Q.2. (Total 4 Marks)

a. You are conducting a food test that requires use of glassware, chemicals and heat. Write any TWO safety measures you need to take during the experiment. (2 Marks)

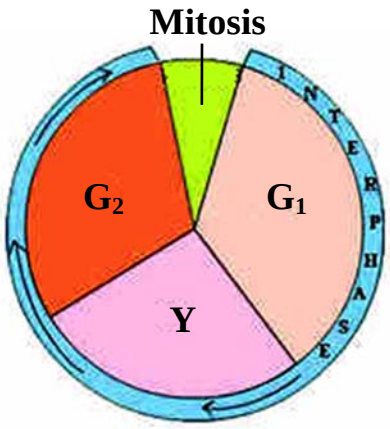
b. Give any TWO possible reasons why a water living organism is endangered? (2 Marks)

Q.3. (Total 4 Marks)

a. The diagram shows different cells of a leaf. Briefly describe how the labelled structures P and Q are specialized to perform their function efficiently. (2 Marks)



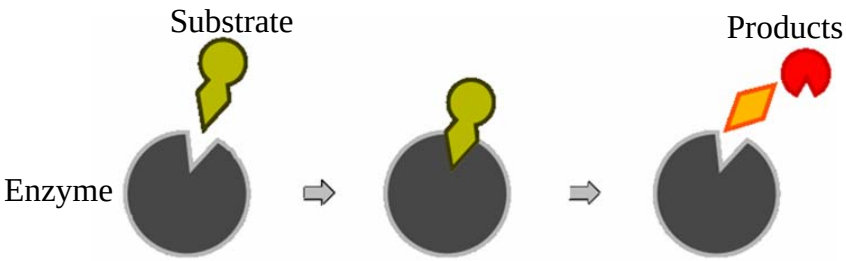
b. The diagram shows a cell cycle. Identify the missing phase Y in the cycle. State what important role it plays in the cell cycle. (2 Marks)



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Q.4. (Total 4 Marks)

a. The diagram shows an enzyme catalysed reaction. (1 Mark)



Which characteristic of enzymes is shown in the diagram?

_____

b. Photosynthesis is a process through which plants prepare their own food. How is this process important for animals? (1 Mark)

_____

_____

c. Give any TWO ways through which respiratory energy is used in a human body. (2 Marks)

_____

_____

_____

_____

Q.5. (Total 4 Marks)

a. The picture below shows the effect of excessive use of fertilizer. (2 Marks)



Give reason for such condition in the plant shown.

_____

_____

_____

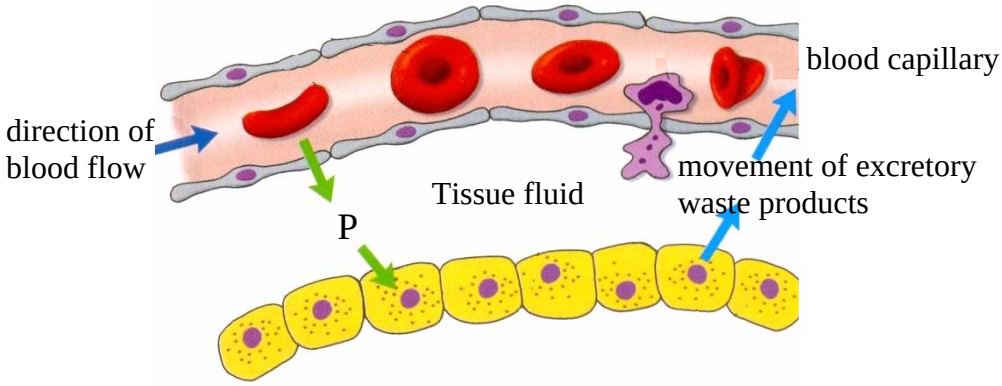
_____

b. The small intestine is about 6-8 metres long. Give TWO reasons of this length being beneficial to the human body? (2 Marks)

Q.6. (Total 5 Marks)

a. Give an account of the effect of increased humidity on the rate of transpiration. (2 Marks)

b. The diagram shows the transference of materials between capillary and tissue fluid.



i. Which structural feature of the blood capillary helps in the transference of materials? (1 Mark)

ii. Name the TWO substances moving from blood capillary to the tissue fluid at region P. (2 Marks)

Q.7. (Total 5 Marks)

EITHER

a. Argue that the five kingdom system of classification of living organisms is better than the two kingdom system. Give examples to support your answer.

OR

b. 'It has now become very important to conserve the population of individual species.' Discuss the above statement indicating the importance of conservation. Give at least THREE examples to support your answer.

Q.8. (Total 5 Marks)

EITHER

a. With the help of labelled diagrams only, show the phenomenon of plasmolysis and deplasmolysis in a plant cell.

OR

b. Describe in detail the sub-phases of the interphase of the cell cycle.

Q.9. (Total 5 Marks)

EITHER

a. What do you understand by the term “energy of activation”? Describe the difference in the activation of energy between catalysed and uncatalysed reactions.

OR

b. Define the term “limiting factor”. How do the following factors affect the rate of photosynthesis?

i. Dim light ii. Low carbon dioxide

END OF PAPER